

Linear Algebra - DS2103

Assignment – 02

01. Solve the following system of linear equation by,

- I. LU factorization method.
- II. Cramer's Rule
- III. Gaussian elimination method
- IV. Gauss-Jordan method

$$x + 2y + 3z = 11$$

$$x - 2y + 3z = 3$$

$$x + 2y - 3z = 11$$

02. Find the eigen values and eigen vectors of the below matrices.

(a) $\begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$ (b) $\begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$ (C) $\begin{bmatrix} 1 & 3 & 3 & -3 \\ 3 & 1 & -3 & 3 \\ 3 & -3 & 1 & 3 \\ -3 & 3 & 3 & 1 \end{bmatrix}$